

GABRIEL WOMELSDORF

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SUMMARY

Computer Engineering student graduating May 2027 with hands-on experience developing embedded firmware and custom hardware using C/C++, Zephyr RTOS/NCS, FreeRTOS, Nordic nRF52/nRF54, STM32, and Bluetooth Low Energy. Builds systems from architecture and PCB design through firmware, sensor integration, power sequencing, validation, and board bring-up; brings 4+ years of production software experience progressing from intern to part-time lead engineer.

EDUCATION

Christopher Newport University

Newport News, VA

B.S., Computer Engineering, *expected May 2027*; Minor, Mathematics

CNU Summer Scholars research grant recipient (*Core4ce-sponsored*), 2026

TECHNICAL SKILLS

Firmware: C, C++, Zephyr RTOS/NCS, FreeRTOS, STM32 HAL, ESP-IDF, Python

Wireless & MCUs: Bluetooth Low Energy (BLE), Nordic nRF52/nRF54, STM32G0

Interfaces & Debug: I2C, SPI, UART, SWD, ADC/DMA, sensor integration

Hardware & Tools: PCB design (KiCad, Altium), board bring-up, USB-C Power Delivery, power sequencing, schematic review, Git, CMake

EMBEDDED SYSTEMS PROJECTS

- **Distributed BLE Sensing System & Production Test Platform** *Capstone, team of two; 2026–Present*
Architected a privacy-preserving multi-node sensing system and designed battery-powered node hardware around Nordic nRF54L15 BLE modules, nPM1300 power management, ADXL367 accelerometers, and SPI flash. Developing a four-layer STM32G0 parallel test fixture with 15-V USB-C PD, protected 5-V/3.3-V rails, 84 test points, and I2C/SPI/UART/SWD. Writing FreeRTOS firmware for PD verification, ADC/DMA rail monitoring, sequenced power-up, fault shutdown, and logging.
- **Air Sentry Neo** *Sep 2025–Present*
Designed and built a battery-powered indoor air-quality research platform end-to-end, including electronics, PCB layout and assembly, enclosure, firmware, five I2C sensors, SPI flash logging, and BLE connectivity. Achieved approximately 3.5 weeks from a 350 mAh LiPo (about 0.6 mA average); owns hardware validation, experimental design, and a manuscript in preparation.
- **Tardis Embedded-AI Development Kit** *Jun–Jul 2026*
Proposed and secured funding for a modular offline edge-AI platform built around an nRF54LM20B system-on-module and AI accelerator. Owned architecture across the compute module, carrier hardware, firmware, and developer tooling; independently assembled and brought up the prototype.

ENGINEERING EXPERIENCE

Core4ce

Reston, VA

Software R&D Intern → Software Engineer → Lead Software Engineer (part-time)

Jun 2022–Present

- Led end-to-end development of Cyberscape, owning architecture decisions, sprint planning, client deployments, and mentorship of junior engineers.
- Designed and implemented a production Apache NiFi data pipeline on AWS, including custom Java processors, IAM-compliant integrations, and reconfigurable routing.
- Led a two-month testing overhaul for a previously untested subsystem, substantially increasing coverage and bringing the component to production-ready quality.

Christopher Newport University

Newport News, VA

Student Researcher (part-time)

Nov 2023–May 2025

- Explored CNC routing methods for fabricating table-sized PCBs for Jefferson Lab's Hall A Super BigBite Coordinate Detector.
- Assembled, installed, and documented power-delivery systems for the detector's photomultipliers.

LEADERSHIP

The Captains Workshop, CNU — Founder and President

Feb 2023–Present

Founded CNU's 3D-printing club and organize hands-on workshops and shared technical resources.